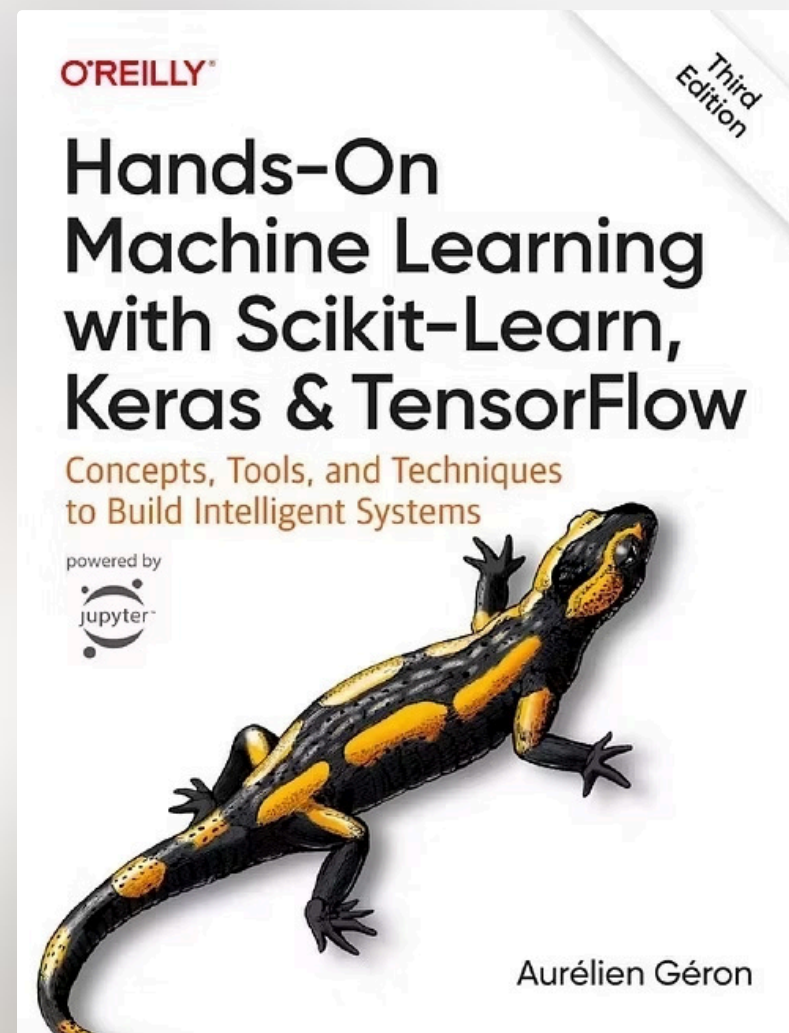


Machine Learning

Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

Last update: 07/04/2025



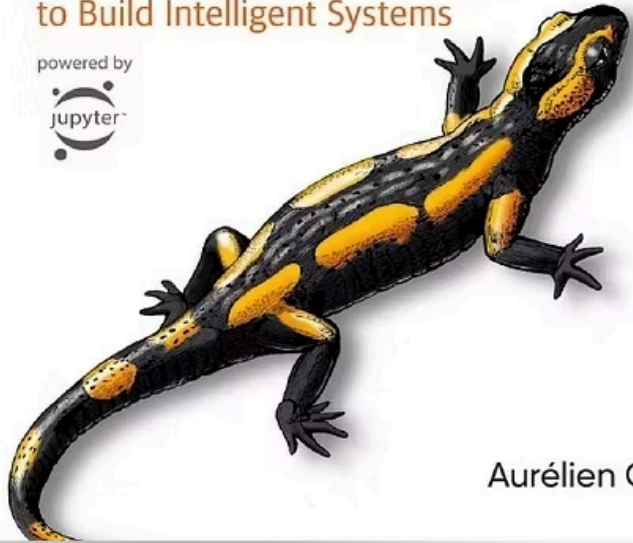
O'REILLY®

Third
Edition

Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow

Concepts, Tools, and Techniques
to Build Intelligent Systems

powered by



Aurélien Géron

Machine Learning

Chapter 1 – The Machine Learning Landscape



Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. **Types of Machine Learning Systems**
5. Main Challenges of Machine Learning
6. Testing and Validating

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. **Types of Machine Learning Systems**
 - **Classification criteria #1:** Supervision during training
 - **Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning**
 - **Classification criteria #2:** Approach to Learning Method
 - Batch and Online Learning
 - **Classification criteria #3:** Learning Method
 - Instance-Based and Model-Based Learning
5. **Main Challenges of Machine Learning**
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - Poor-Quality Data
 - Irrelevant Features
 - Overfitting the Training Data
 - Underfitting the Training Data
 - Stepping Back
6. **Testing and Validating**
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch



Classification criteria #1: Supervision during training

Supervised, Unsupervised, **Semi-Supervised**, Self-Supervised, and Reinforcement Learning

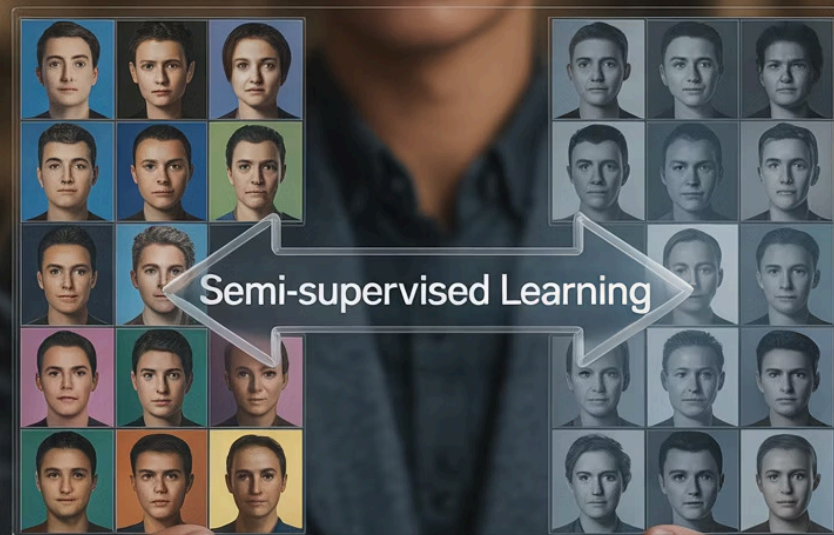


Since labeling data is usually time-consuming and costly, you will often have plenty of unlabeled instances, and few labeled instances

Semi-Supervised Learning

Combines a small amount of labeled data with a large amount of unlabeled data.

Example: Recognizing faces in photos with a few labeled instances and many unlabeled ones.



Semi-Supervised Learning example

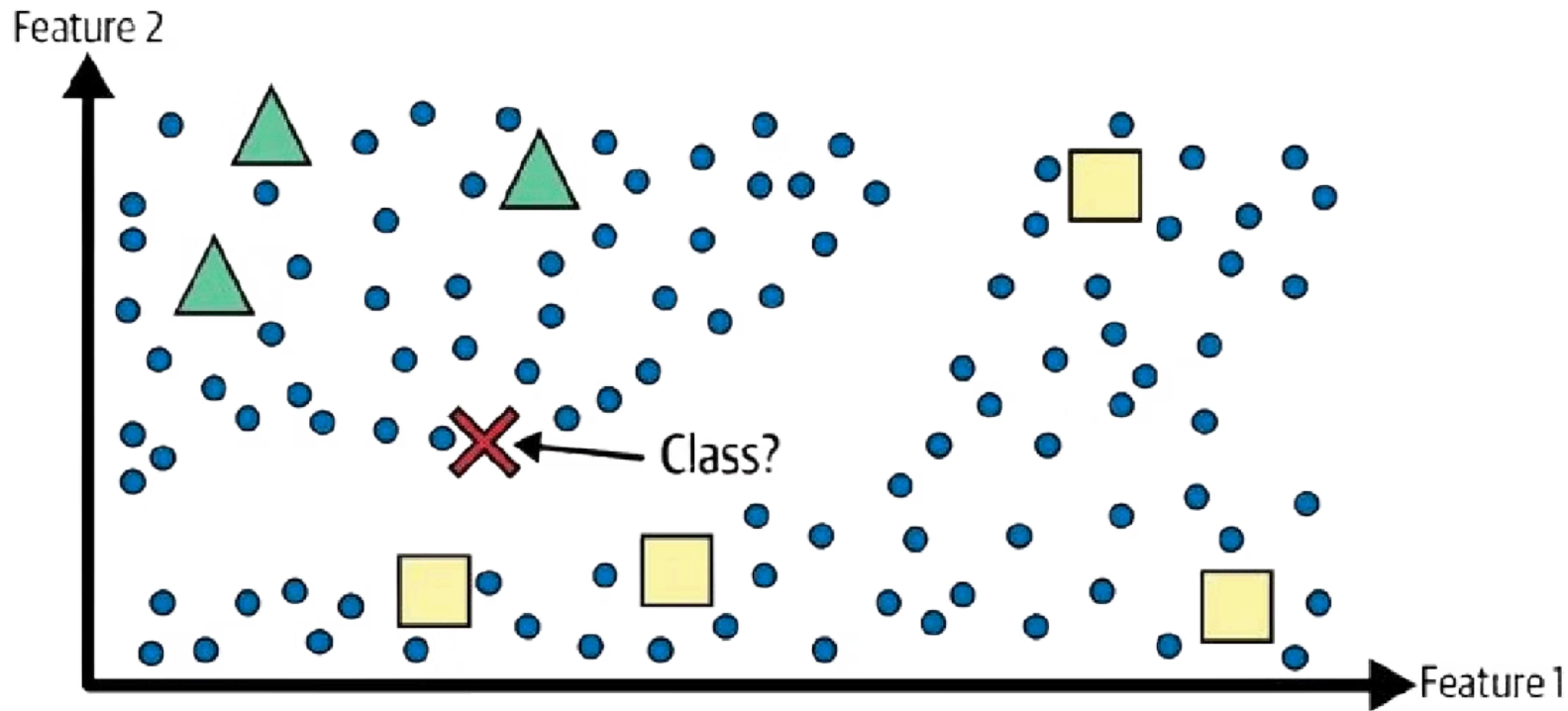
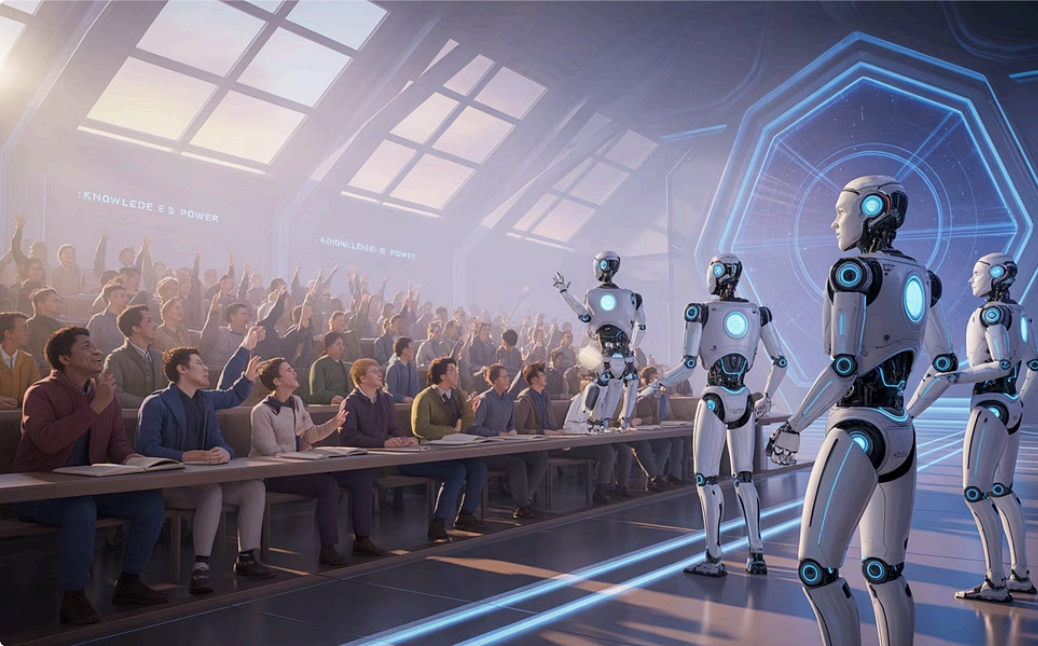


Figure 1-11. **Semi-supervised learning** with two classes (**triangles** and **squares**): the unlabeled examples (**circles**) help classify a new instance (**the cross**) into the **triangle** class rather than the **square** class, even though it is closer to the labeled squares

Next: Self-Supervised Learning

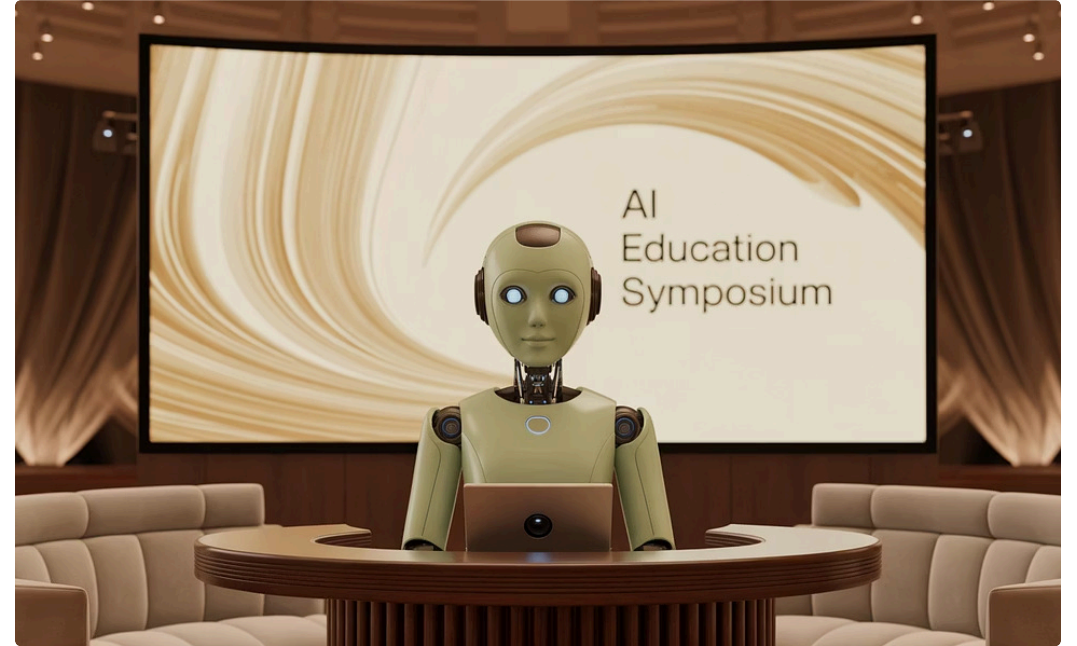


Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts